

Applied Macroeconomics
Course overview
Introduction
Two-period consumption-saving model

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Course overview

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 - Zoom consultation link: see Moodle

Lecture schedule

The program outlined below is a guide to the material to be covered in this unit and not a definitive statement of what and when that material will be covered.

- **Topic 1.** Consumer's intertemporal consumption-savings decision: Weeks 1, 2. Reading: Chapter 4 the representative consumer, Chapter 9.
- **Topic 2.** Credit market imperfections: Week 3. Reading: Chapter 10
- **Topic 3.** Consumer's intertemporal consumption-savings and work-leisure decisions: Week 4. Reading: Chapter 11 the representative consumer.
- **Topic 4.** Firm's intertemporal problem: Week 5. Reading: Chapters 4 and 11 the representative firm.

Lecture schedule (continue)

- **Topic 5.** [Real intertemporal model with investment](#): Week 6.
Reading: Chapter 11
- **Topic 6.** [Monetary intertemporal macro model](#): Week 7. Reading:
Chapter 12
- **Topic 7.** [Business Cycle Models with Flexible Prices and Wages](#):
Week 8. Reading: Chapter 13
- **Topic 8:** [Business Cycle Model with Sticky Prices](#): Week 9. Reading:
Chapter 14

Lecture schedule (continue)

- **Topic 9:** [Search and unemployment](#): Week 10. Reading: Chapter 6.
- **Topic 10.** [Economic Growth: Malthus](#): Week 11. Reading: Chapter 7.
- **Topic 11.** [Economic Growth: Solow](#): Weeks 11, 12. Reading: Chapter 7; Blanchard and Sheen, sections 12.1-12.4.
- **Topic 12.** [Endogenous Growth](#): Week 12. Reading: Chapter 8.

- Prescribed text:
 - Macroeconomics by Williamson, S., 6e, Global Edition, Pearson.

Assessments

Assessment task	Value	Due date
Online Mid-semester Test (70 minutes)	30%	Week 6 on 8 September 2020 (Tuesday), starts at 6pm Melbourne time (any change will be announced at Moodle)
Submit 4 Problem sets: 2% each	8%	Submit online at Moodle on Tuesday by 6pm Melbourne time in Week 3, Week 5, Week 8, Week 10
Project report (max. 12 pages)	12%	Submit online at Moodle in Week 11, Tuesday by 6pm Melbourne time
Examination	50%	To be advised

Within semester assessment

- The mid-semester test relates to materials covered in the first five weeks (including week 5).
- For the individual project report, see Moodle for details (the project topics, writing guide, criteria for marking).
- Please check your Monash email and Moodle everyday.

Learning outcomes

- Analyse macroeconomic issues using formal language
- Understand different theories that can be used to analyse real world macroeconomic problems and government policies
- Develop critical thinking skills
- Prepare for advanced study in macroeconomics.

- Macroeconomics deals with the entire (aggregate) economy.
- In macroeconomics, we build economic models, which
 - embody the theory that we use to understand problems of interest
 - are simple enough that the key mechanisms that make it work can be easily understood
 - help us gain insight about the workings of the vast macroeconomy
 - take exogenous variables and explain the endogenous variables:
 - Exogenous: something that is given from outside the model

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 - Endogenous: something that is determined within the model

Macroeconomics in this unit

- This unit follows a modern approach by building macro models from microeconomic principles, i.e. study what a consumer or firm does.
 - Aggregate consumption is the aggregation of individual's consumption decisions
 - Aggregate investment is the aggregate of individual firms' investment decisions.
- This modern approach has 3 advantages:
 - It is consistent with the approach that macro research is conducted today.
 - It allows deeper insights into key topics in macroeconomics.
 - It allows students to be better prepared for advanced study in economics.

Macroeconomics in this unit

- We will study specific models, start from a simple model extend to include more sectors (from consumers, firms to government), to explain how output, employment and the price level are determined.
- Study the general equilibrium approach to analyze business cycles and the effects of monetary and fiscal policies.
- Apply theories to analyze current macroeconomic issues.
- Study long run economic growth in a dynamic setting.

Two-Period Consumption-Saving Model

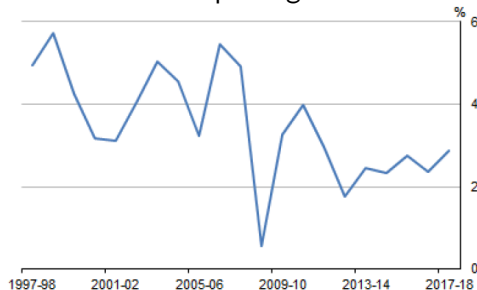
- Consumer's intertemporal consumption-savings decision
- Reading: Williamson Chapters 4, 9

Consumption and savings

- Recall: **Gross Domestic Product (GDP)**: a measure of aggregate output (all final goods and services) produced within a country's borders over a particular period of time.
- GDP can be determined using the **income-expenditure identity**:
 - Total income = total spending: $Y = C + I + G + NX$
- **Consumption**: spending by consumers (or household).
- C accounts for more than half of GDP. Example: C in Australia is 54% of GDP in 2011.
- Consumption decision is closely linked to saving decision: household savings are disposable income that is not consumed.

Consumption and savings

- Household consumption growth in Australia:



Source: Australian Bureau of Statistics (ABS)

- What causes consumption to change? Factors affecting consumption as well as saving decisions.

Representative consumer

- Assume all consumers are identical (homogenous), so having consumers who are identical is identical to having one representative consumer.
- Assume the consumer lives in two periods, period 1 (the current period, or today) and period 2 (the future period, or tomorrow). All variables are denoted in terms of consumption goods in this model.
- The consumer's current-period budget constraint:

$$c + s = y - t \quad (1)$$

- The consumer's future-period budget constraint:

$$c' = y' - t' + (1 + r) s \quad (2)$$

- y (y') is the current (future) income, t (t') is the current (future) tax, s is saving, and r is the real interest rate. Assume exogenous income.

Representative consumer

- Solve the future-period budget constraint for s :

$$s = \frac{c' - y' + t'}{1 + r} \quad (3)$$

- Saving can be positive (lending) or negative (borrowing), or 0 (no lending or borrowing).
- If the consumer chooses not to consume his entire income in the first period, his saving is positive, $s > 0$, which earn a real rate of interest r . In the second period, he gets back principal plus interest, $(1 + r)s$.
- If he chooses to borrow so as to consume more than his income in the first period, then $s < 0$; and in the second period he pays back $(1 + r)s$.

Representative consumer

- Think of this as a credit market, where an individual can issue bonds if he wants to borrow, or buy bonds from other individuals if he wants to save.
- Assume perfect credit market: no default, everyone issues bonds and buys bonds at the same rate r .
- Assume all bonds are traded directly on credit market; no banks.

Representative consumer

- Saving 1 unit of consumption good in the current period gives you $(1 + r)$ units of consumption good in the future period.
 - The relative price of current consumption in terms of future consumption is $(1 + r)$: one unit in the current period is worth $(1 + r)$ units in the future.
 - The relative price of future consumption in terms of current consumption is $1/(1 + r)$: one unit in the future is worth $1/(1 + r)$ units in the current period.
- When denoting future values by how much it is worth in the current period, it is called **PRESENT VALUE**.

Representative consumer

- Substitute (3) in the current-period budget constraint obtaining lifetime budget constraint:

$$c + \frac{c'}{1+r} = y + \frac{y'}{1+r} - t - \frac{t'}{1+r}$$

- The present value of lifetime consumption (the left hand side) must equal the present value of lifetime disposable income (the right-hand side).

Representative consumer

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- The present value of lifetime consumption (the left hand side) must equal the present value of lifetime disposable income (the right-hand side).
- Define the consumer's lifetime wealth, we , to be the present value of lifetime disposable income:

$$we = y + \frac{y'}{1+r} - t - \frac{t'}{1+r}$$

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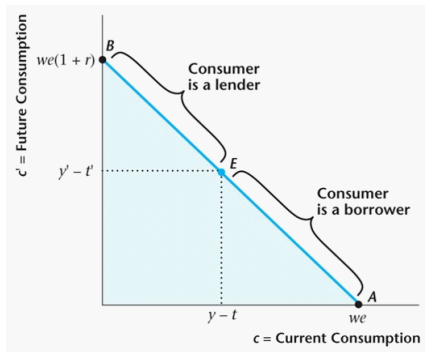
$$we = y + \frac{y'}{1+r} - t - \frac{t'}{1+r}$$

- The simplified lifetime budget constraint:

$$c + \frac{c'}{1+r} = we \Rightarrow c' = -(1+r)c + we(1+r)$$

Representative consumer

- The endowment point $E (y - t, y' - t')$, if consumed, implies that the consumer would save zero in the first period, and consume his disposable income in each period. To the northwest of E he is a lender, and to the southeast, he is a borrower.



Representative consumer

- The consumer's preferences are given by the lifetime utility function $u(c, c')$, where c is current consumption and c' is future consumption. c and c' are normal goods.
- The utility function is strictly increasing and strictly concave:
 - Marginal utility (MU) of consumption is positive (more is better): $\partial u / \partial c > 0, \partial u / \partial c' > 0$.

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- The utility function is strictly increasing and strictly concave:
 - Marginal utility (MU) of consumption is positive (more is better): $\partial u / \partial c > 0$, $\partial u / \partial c' > 0$.
 - Diminishing MU (as consumption gets higher, MU from more consumption gets smaller): $\partial^2 u / \partial c^2 < 0$, $\partial^2 u / \partial (c')^2 < 0$.

Representative consumer

- The consumer maximizes his lifetime utility, $u(c, c')$, subject to his lifetime budget constraint (LBC), $c' = -(1+r)c + we(1+r)$.
- Substituting the LBC into the lifetime utility (or use the Lagrangian method):

$$\max_{c, c'} u(c, -(1+r)c + we(1+r))$$

The FOCs:

$$\frac{\partial u}{\partial c} - \frac{\partial u}{\partial c'}(1+r) = 0$$

$$\frac{\partial u / \partial c}{\partial u / \partial c'} = MRS_{c, c'} = 1 + r$$

The rate at which the consumer is willing to trade the current consumption for future consumption is equal to the rate at which the consumer is able to trade.

Representative consumer

- Consumer's preferences over current and future consumption are represented graphically by indifference curves (IC).
- An IC
 - connects a set of points, representing consumption bundles among which the consumer is indifferent

Representative consumer

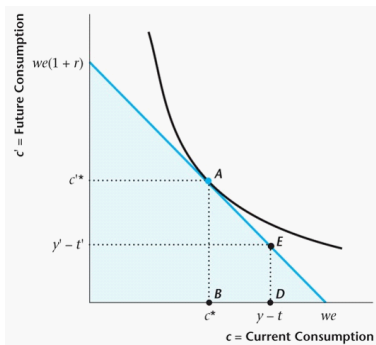
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Representative consumer

- Consumer's preferences over current and future consumption are represented graphically by indifference curves (IC).
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 - connects a set of points, representing consumption bundles among which the consumer is indifferent
 - it slopes downward: more is preferred to less
 - it is convex: preference for diversity in consumption bundle

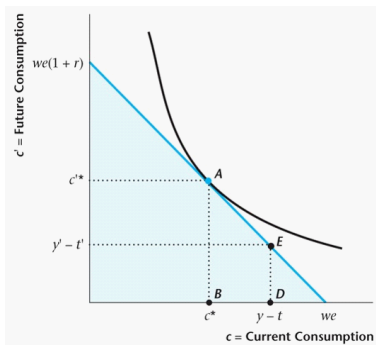
Representative consumer

- The FOC gives the tangency of IC to LBC, which picks out the point of the maximum utility a consumer can obtain given his LBC.
 - The slope of the IC (in absolute value) is $MRS_{c,c'}$ which is $(\partial u / \partial c) / (\partial u / \partial c')$.



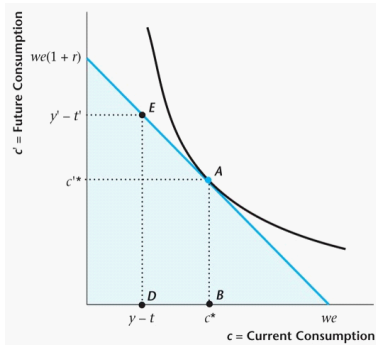
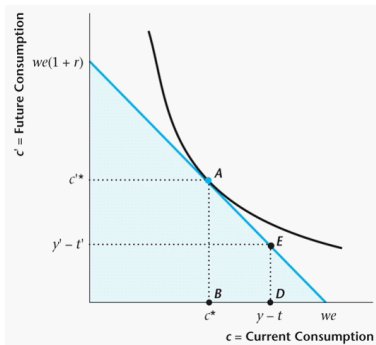
Representative consumer

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 - The slope of the IC (in absolute value) is $MRS_{c,c'}$ which is $(\partial u / \partial c) / (\partial u / \partial c')$.
 - The slope of the LBC (in absolute value) is $1 + r$.



Representative consumer

- If the optimization point is above (below) point E along the LBC, then this consumer is a lender (borrower).



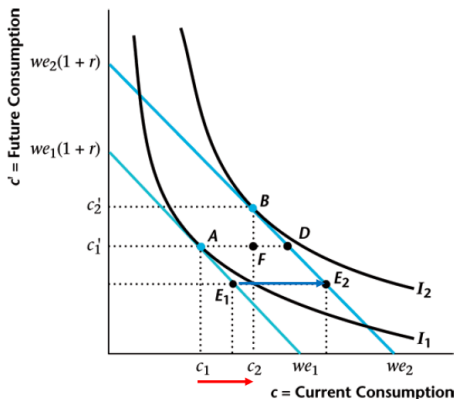
- What is the key message to take out of consumer's maximization problem thus far?
 - Consumers consumes out of lifetime wealth, not just current income
- Since an economy is composed of many, many agents, each agent is making optimal consumption choices each period based on what their lifetime wealth is. Aggregate consumption is thus an aggregation of each individual's consumption choices.

Comparative Statics: Increase in Current Income

- Suppose current income y increases.
- Then lifetime wealth increases by $\Delta y = we_2 - we_1$. So the LBC shifts to the right by Δy .
- Since the interest rate is unchanged, the slope of LBC remains constant.
- With this pure income effect, both c and c' increase since they are normal goods.
- Saving $s = y - t - c$ increases as $\Delta c < \Delta y$.
- This reflects **consumption smoothing**: consumer is "spreading out" increase in current income over both periods.

Comparative Statics: Increase in Current Income

- When current income increases, the optimal allocation moves from A to B

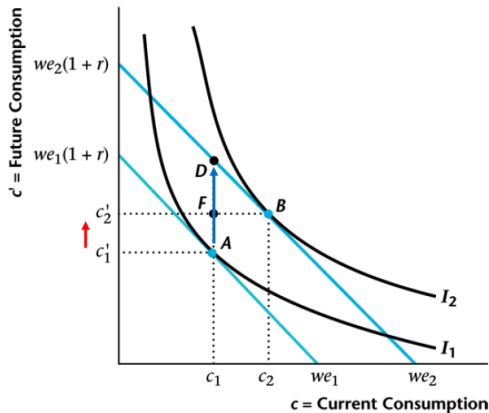


Comparative Statics: Increase in Future Income

- Suppose future income y' increases.
- Then lifetime wealth increases from we_1 to we_2 , shifting the LBC to the right by $we_2 - we_1$.
- So the LBC shifts up by $\Delta y' = (1 + r)(we_2 - we_1)$.
- Since the interest rate is unchanged, the slope of LBC remains constant.
- A pure income effect: both c and c' increase, while $s = y - t - c$ decreases.

Comparative Statics: Increase in Future Income

- When future income increases, the optimal allocation moves from A to B



Comparative Statics: Increase in Income

Temporary vs. permanent changes in income

- A permanent increase in income will have a larger effect on lifetime wealth than a temporary increase. So, there will be a larger effect on current consumption.
- A consumer will tend to save most of a purely temporary income increase.
- **Permanent Income Hypothesis** (Milton Friedman)
 - Current consumption depends on lifetime wealth (closely related to permanent income). Changes in temporary income yield smaller changes in lifetime wealth (permanent income) which affects current consumption by less than permanent changes in income (which yield larger changes in lifetime wealth, and current consumption).

Comparative Statics: Increase in Income

Temporary vs. permanent changes in income

- Increase in y is a temporary change in income. LBC shifts from AB to DE.
 - The optimal allocation moves from H to J. Positive saving.
- Increase in both y and y' is a permanent change in income. LBC shifts from AB to FG.

Comparative Statics: Increase in Income

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- Increase in y is a temporary change in income. LBC shifts from AB to DE.
 - The optimal allocation moves from H to J. Positive saving.
- Increase in both y and y' is a permanent change in income. LBC shifts from AB to FG.
 - The optimal allocation moves from H to K. The effect on saving is ambiguous (recall: increase in y' cause s to fall, ceteris paribus).

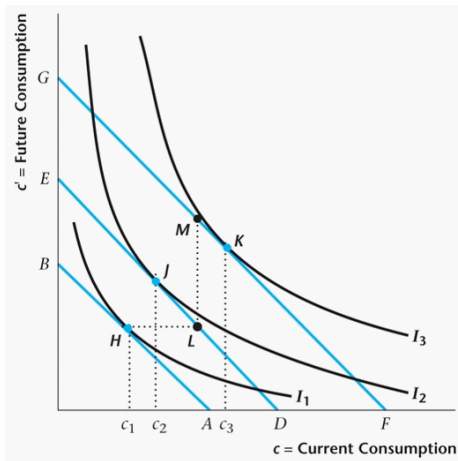
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 - Larger effect on current consumption.

Comparative Statics: Increase in Income

Temporary vs. permanent changes in income



Comparative Statics: Increase in Income

Temporary vs. permanent changes in income

- Policy implication
 - This matters for tax policy, for example.

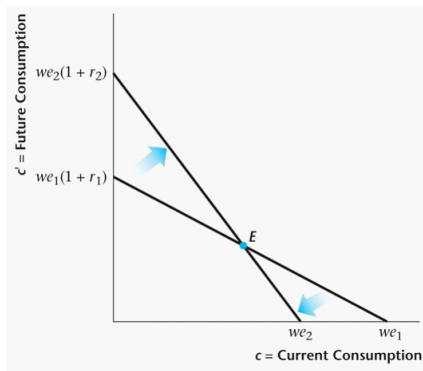
Comparative Statics: Increase in Income

Temporary vs. permanent changes in income

- Policy implication
 - This matters for tax policy, for example.
 - A permanent tax cut will in principle imply a larger effect on consumption spending than a temporary tax cut.

Comparative Statics: Increase in the Real Interest Rate

- This changes the relative price of future consumption goods in terms of current consumption goods.
- $w e_2$ decreases and $w e_1 (1 + r)$ increases. The new LBC is steeper than the old one, and goes through the endowment point since a consumption plan $(y - t, y' - t')$ here is independent of the level of the real interest rate.



Comparative Statics: Increase in the Real Interest Rate

- This has income effect (IE) and substitution effect (SE).
- The SE: c is more expensive and c' is cheaper, thus c decreases and c' increases. Saving s will increase.
- The IE depends on whether the consumer is a lender or a borrower.
 - For a lender, a higher interest rate implies that the future income increases, leading to a higher c and c' , but lower s .

Comparative Statics: Increase in the Real Interest Rate

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- The SE: c is more expensive and c' is cheaper, thus c decreases and c' increases. Saving s will increase.
- The IE depends on whether the consumer is a lender or a borrower.
 - For a lender, a higher interest rate implies that the future income increases, leading to a higher c and c' , but lower s .
 - For a borrower, a higher interest rate reduces future income and hence c and c' . Saving s will increase.

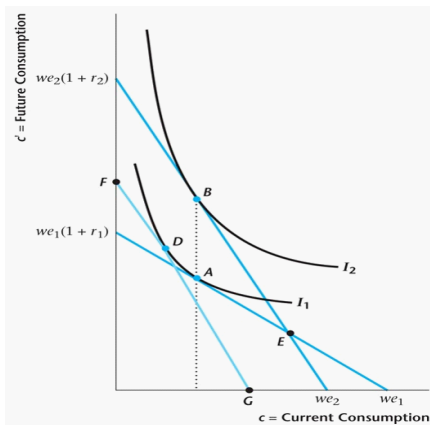
Comparative Statics: Increase in the Real Interest Rate

- The aggregate effect.
 - For a lender, future consumption c' increases, but both c and s are ambiguous.

Comparative Statics: Increase in the Real Interest Rate

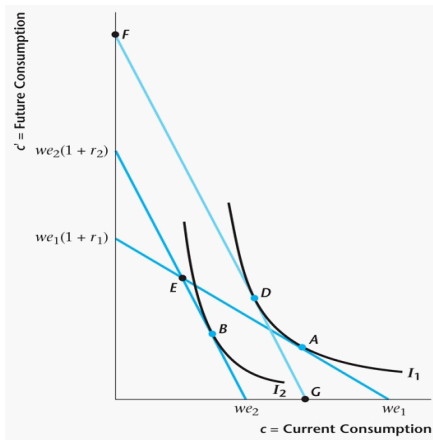
- The aggregate effect.
 - For a lender, future consumption c' increases, but both c and s are ambiguous.
 - For a borrower, c falls and s increases, but future consumption c' is ambiguous.

Increase in the Real Interest Rate for a Lender



- Starting point: E
- Optimal consumption: A with r_1
- Budget constraint becomes steeper, going through E
- Optimal consumption: B with r_2
- A to D is the substitution effect
- D to B is the income effect
- As a result, future consumption increases
- Current consumption and savings may rise or fall

Increase in the Real Interest Rate for a Borrower



- Starting point: E
- Optimal consumption: A with r_1
- Budget constraint becomes steeper, going through E
- Optimal consumption: B with r_2
- A to D is the substitution effect
- D to B is the income effect
- As a result, current consumption falls
- savings increases

- Denote S^g as government saving or surplus, T as taxes, TR as transfers to the private sector, INT as interest on the government debt, and G as government spending.
- Then, $S^g = T - TR - INT - G$
- Since $-S^g$ is the government deficit which is simply government borrowing. So, $-S^g = B$.
- Suppose $TR = 0$ and $INT = 0$, and the government finances G by taxing T , and borrowing B in the current period. Then, the government's current-period budget constraint:

$$G = T + B$$

- In the future period, the government spends and pays off its debt by taxing. There is no bond issued, as this is the last period.

$$G' + (1 + r) B = T'$$

- Substituting out for B using the two government budget constraints, the government's present-value budget constraint is

$$G + \frac{G'}{1 + r} = T + \frac{T'}{1 + r}$$

which says the present value of government spending must equal the present value of taxes, assuming the government never defaults on its debt.

Ricardian Equivalence

- This theorem shows that, under ideal conditions, deficits are irrelevant, i.e. the timing of taxes is irrelevant.
- This theorem however helps us understand why government deficits matter.
- **The theorem of Ricardian Equivalence** : If current and future government spending are held constant, then a change in current taxes with an equal and opposite change in the present value of future taxes leaves the equilibrium interest rate and the consumption of individuals unchanged.

Ricardian Equivalence

- Suppose there are N consumers and, assuming each consumer pays the same taxes in the present and the future.
- Recall that each consumer faces the following LBC

$$c + \frac{c'}{1+r} = y + \frac{y'}{1+r} - t - \frac{t'}{1+r}$$

where $t + t'/(1+r)$ is the consumer's lifetime tax burden.

- The government's present value budget constraint is the present value of government spending equals to the present value of taxes received by the government:

$$G + \frac{G'}{1+r} = T + \frac{T'}{1+r} = N \left(t + \frac{t'}{1+r} \right)$$

Ricardian Equivalence

- Key equation: The consumer's lifetime tax burden is equal to the consumer's share of the present value of government spending – the timing of taxation (present vs. future) does not matter for the consumer:

$$t + \frac{t'}{1+r} = \frac{1}{N} \left(G + \frac{G'}{1+r} \right)$$

- The consumer's LBC is unaffected by whether current taxes are higher or lower:

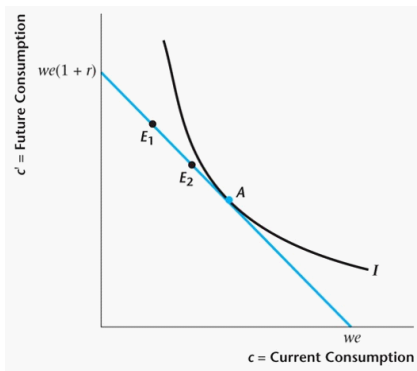
$$c + \frac{c'}{1+r} = y + \frac{y'}{1+r} - \frac{1}{N} \left(G + \frac{G'}{1+r} \right)$$

In equilibrium, the present value of consumption depends only on the present value of government spending – not on taxes.

Ricardian Equivalence: A Cut in Current Taxes

- Given G and G' unchanged, if the government cuts each consumer's current tax t by 1 unit and finances its budget deficit by borrowing 1 unit, the government must increase each consumer's future taxes t' by $(1 + r)$ units to pay off the debt plus interest ($1 + r$ units per consumer).
- A current tax cut moves the endowment point, but does not change the LBC for the consumer. The consumer consumes the same amounts. He saves the tax cut (1 unit) to pay the higher taxes in the future ($1 + r$ units). These higher future taxes are required to pay off the higher government debt.
- Thus, the change in government dissavings equals the change in private savings.

Ricardian Equivalence: a Cut in Current Taxes for a Borrower



Application: Why Might Ricardian Equivalence Fail in Practice?

- Ricardian Equivalence result relies on several assumptions. When some of these assumptions are relaxed, it can fail. Four instances under which the Ricardian Equivalence can fail:
 - Intragenerational Transfers: tax changes affect the wealth of different consumers in the same generation differently.

Application: Why Might Ricardian Equivalence Fail in Practice?

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 - Distortionary Taxes: taxes are not lump sum; they cause distortions.
 - Credit Market Imperfections: credit constraint.

Application: Why Might Ricardian Equivalence Fail in Practice?

- Intragenerational Transfers
 - Within each generation, tax cuts unevenly distributed, and subsequent tax increase unevenly distributed. In other words, it could be that only some people get a tax cut, but others pay for it in second period. In this case, an individual's LBC changes, because his wealth is different, so optimal consumption choices are different.

Application: Why Might Ricardian Equivalence Fail in Practice?

- Intergenerational Transfers
 - Consider a four-period economy with two generations who each live two periods. First generation born in period one and second generation born in period three.

Application: Why Might Ricardian Equivalence Fail in Practice?

- Intergenerational Transfers

- Consider a four-period economy with two generations who each live two periods. First generation born in period one and second generation born in period three.
- If the government cuts taxes in period one and increases taxes in period three to pay off its debt plus interest, i.e., if there is income redistribution from generation 1 to generation 2, then
 - first generation has an increase in their lifetime wealth, so their consumption (in period one and period two) increases
 - second generation has a decrease in lifetime wealth, so their consumption (in period three and period four) decreases

Application: Why Might Ricardian Equivalence Fail in Practice?

- Intergenerational Transfers: Ricardian Equivalence can be restored by parental altruism
 - Suppose the parent benefits from a tax cut and his children will have an increase in taxes in future. The altruistic parent will save more today to give his children more bequests to offset the increase in taxes they will need to pay in future. Hence, there will be no change in the parent's consumption and no change in his children's consumption following the change in the timing of taxes.

Application: Why Might Ricardian Equivalence Fail in Practice?

- Distortionary Taxes
 - If proportional taxes, i.e., $t = \theta y$, this distorts labour supply choice: changes in timing of taxes affect labour supply choices, so there is no longer Ricardian Equivalence.
- Credit constraint

Application: Why Might Ricardian Equivalence Fail in Practice?

- Distortionary Taxes

- If proportional taxes, i.e., $t = \theta y$, this distorts labour supply choice: changes in timing of taxes affect labour supply choices, so there is no longer Ricardian Equivalence.

- Credit constraint

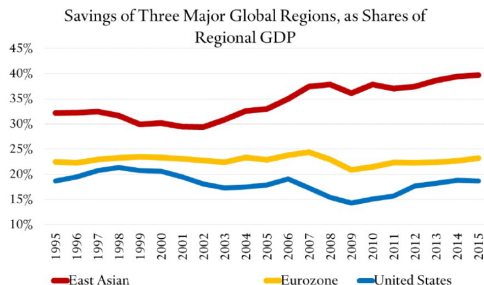
- If some individuals would like to consume more today but cannot because they cannot borrow more (i.e., they are credit constrained), then if they are given a tax cut today, they will consume more today and consume less tomorrow.

Application: Precautionary savings

- So far we have been assuming that the consumer faces no uncertainty.
- How would the consumer's decisions change if he/she faces uncertainty?
- Suppose second period income is uncertain; can be $y' - \epsilon$ with probability 0.5, and $y' + \epsilon$ with probability 0.5; $\epsilon > 0$.
- On average, the consumer makes the same income in the second period. Does this mean he/she will make the same consumption choices as he/she faces no uncertainty?
- Risk-averse people put more weight on the bad state, which make them save more and consume less. This effect of uncertainty on consumption behavior is known as "precautionary savings".

Application: Precautionary savings

- Recall: we assume diminishing marginal utility, which results in risk aversion (can prove this mathematically but it's not required here).
- Precautionary savings may explain why savings are high in some countries. For example, Bernanke (2005) identifies the precautionary motive as the underlying source of high Asian (China, Japan, South Korea, Taiwan, Hong Kong, Singapore) saving rates:



Source: IMF